FIRST SHELL; LINUX 101

U3VwY29tIEN5YmVyc2VjdXJpdHkgQ2x1Yg

Operating System ?

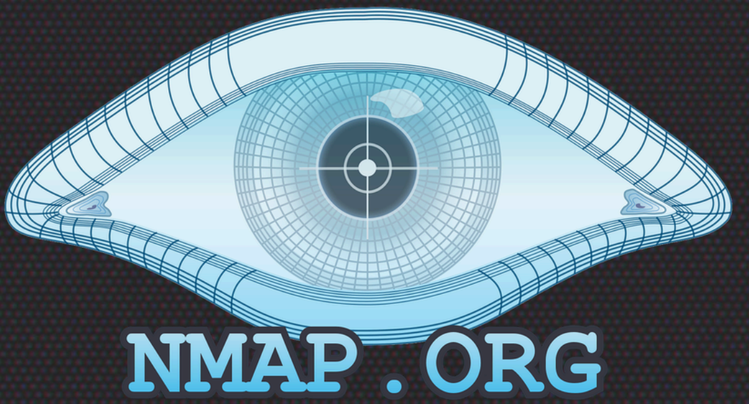
- Runs hardware
- Manage resources
- Provides security
- Organise files

UNIX®

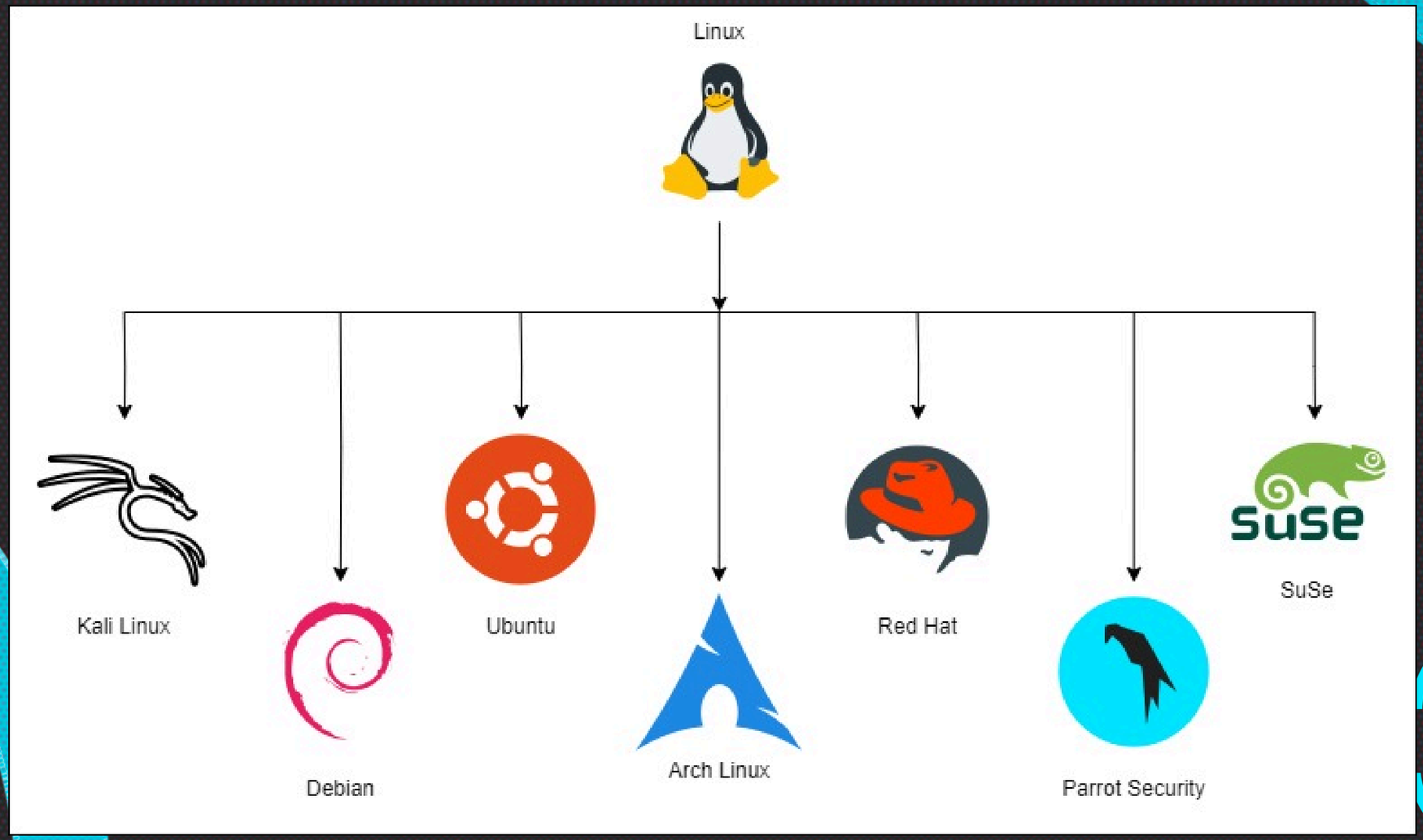
Mac OS

Why Linux ?

- Absolute control
- Free, Secure and Open Source
- Built for security tools

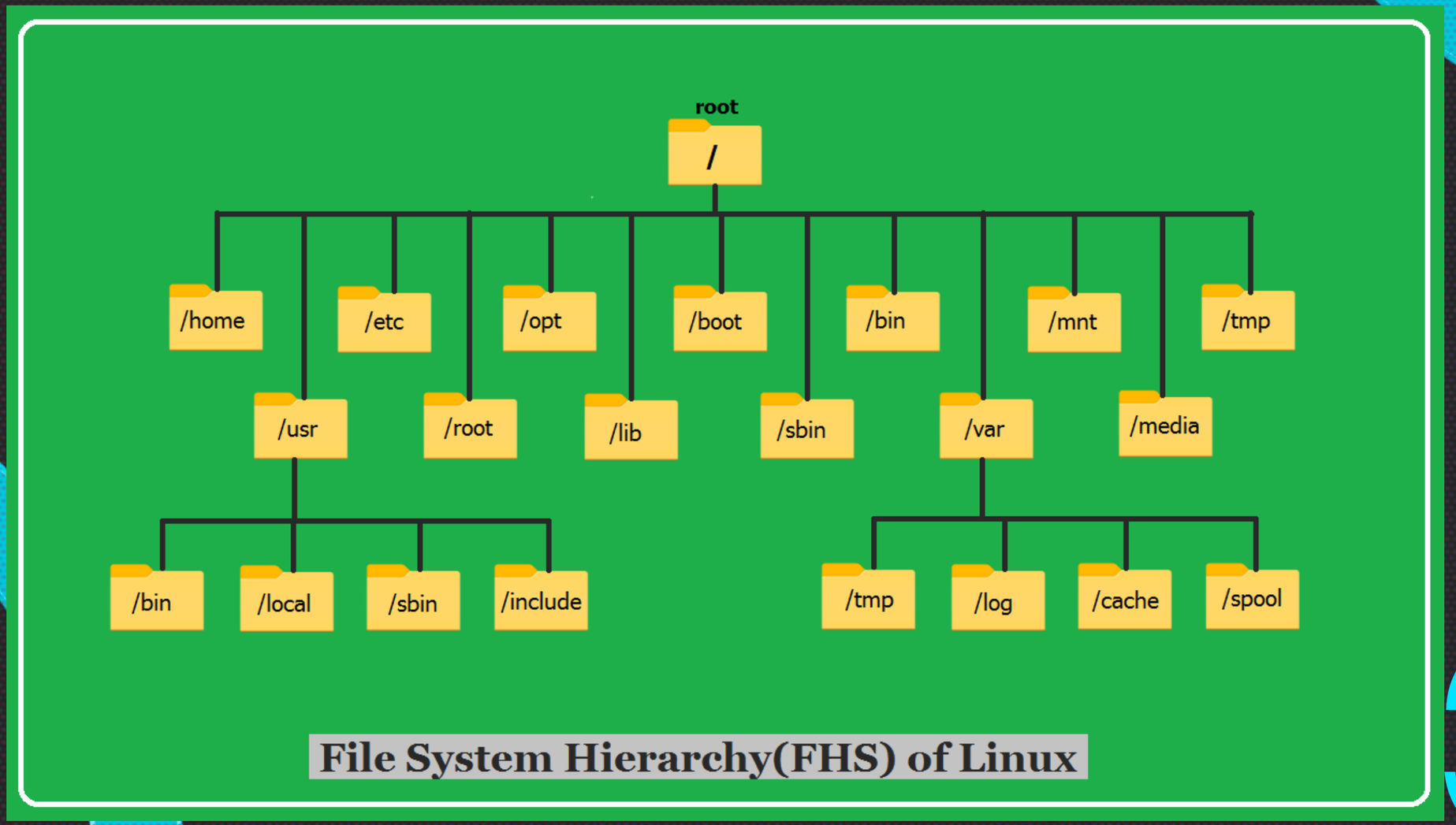


Linux Distros

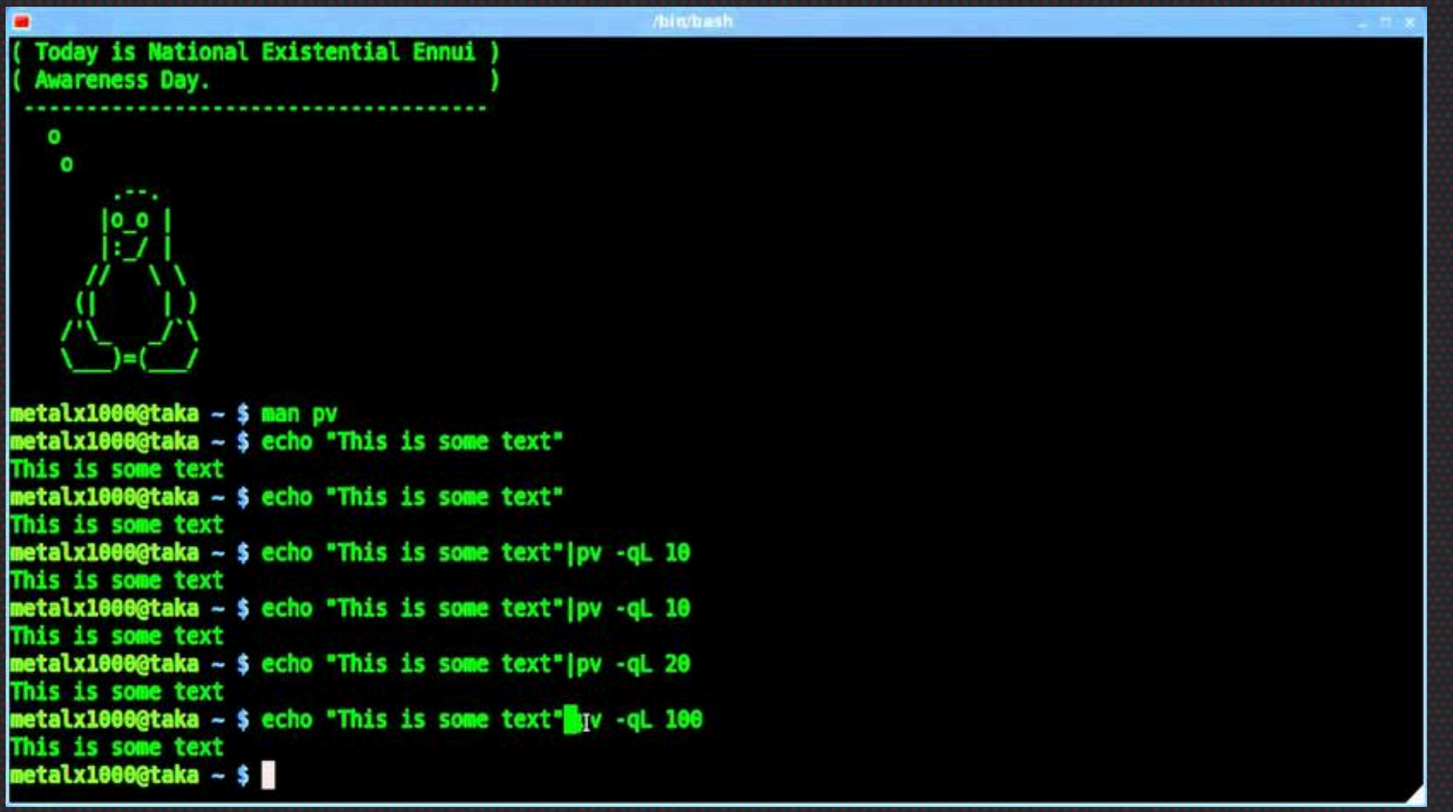


Windows Directory Tree

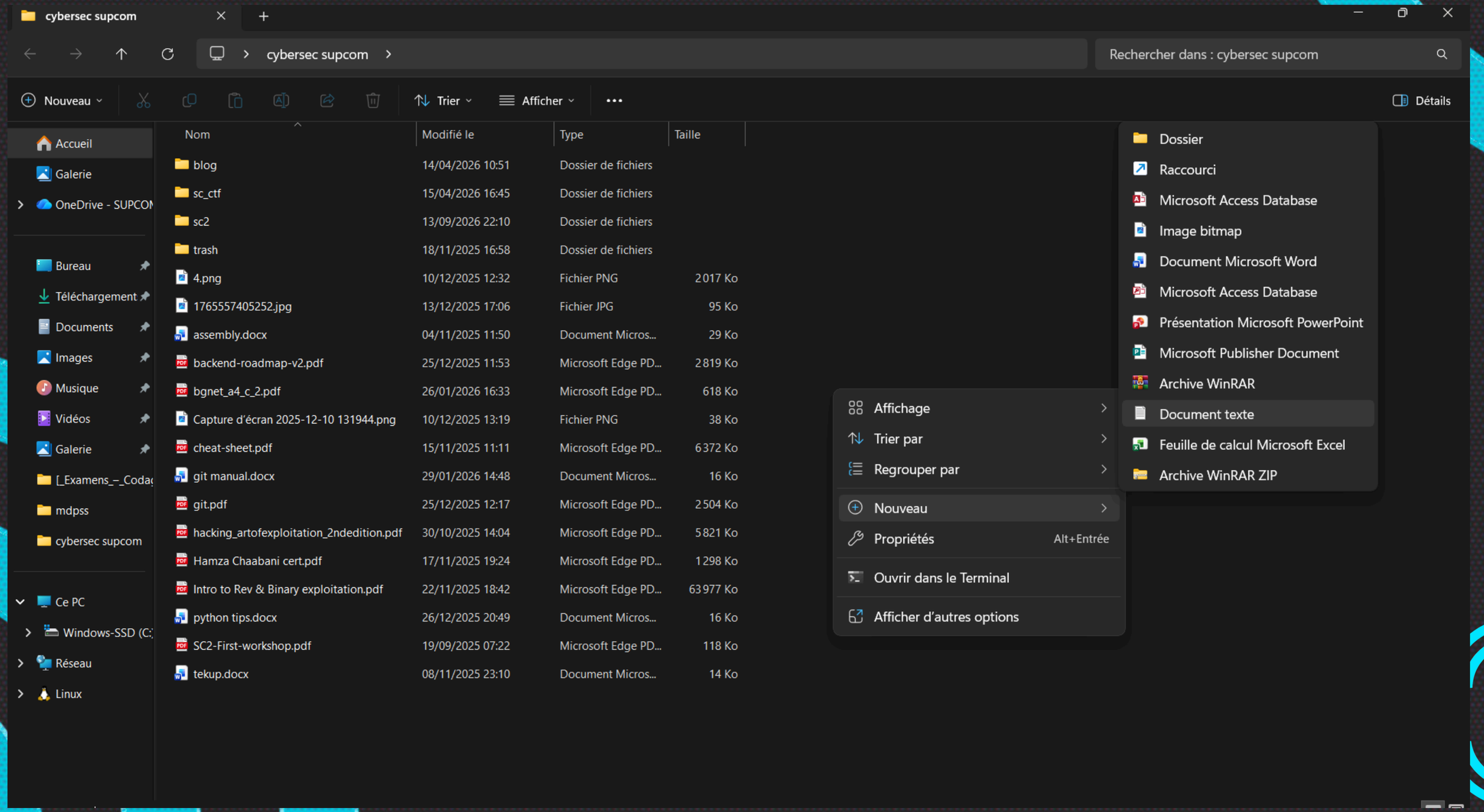
Linux File System



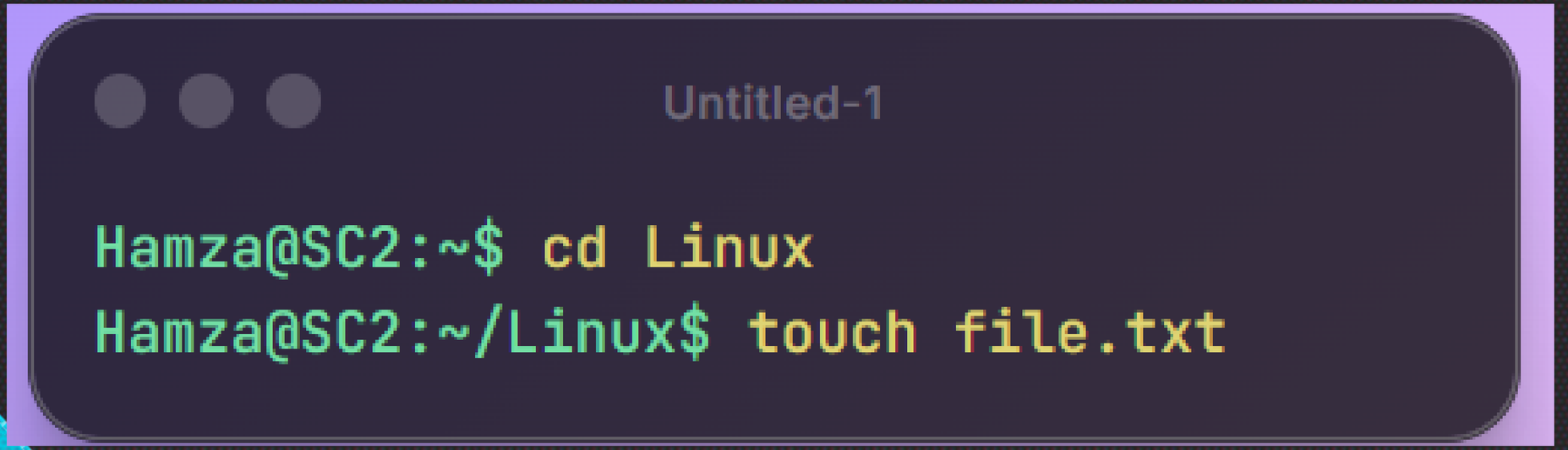
What is a shell ?



Creating a file in windows



Creating a file with the shell

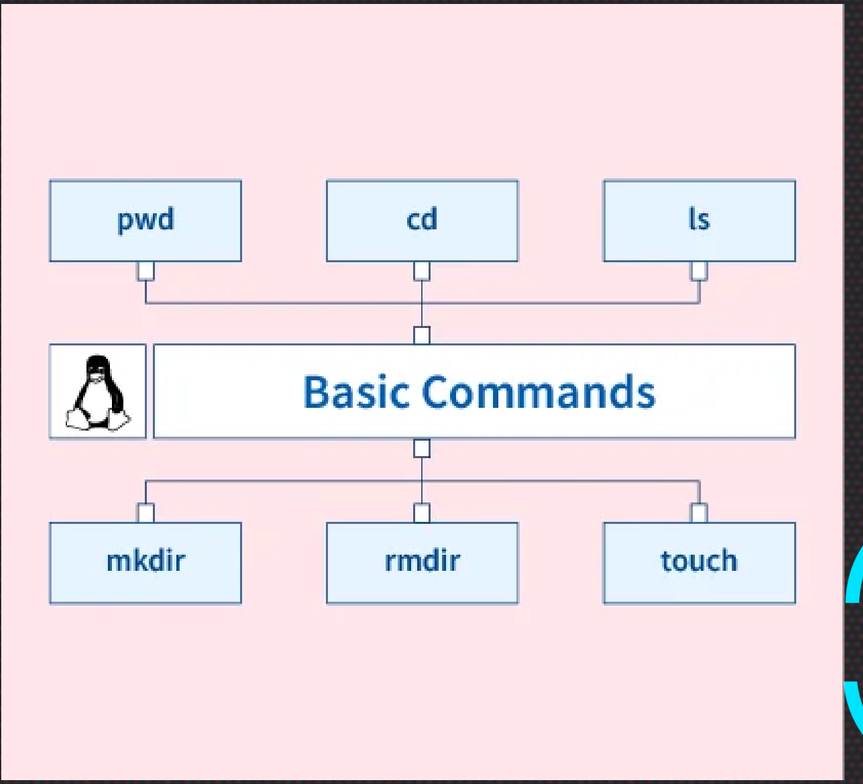


Shell Commands

Format: command name, argument and option/flag

<**Verb**> <**Modifier**> <**Target**>

mkdir linux
cd linux
touch test.txt
ls /home
ls -l /home



Practice

find your current directory
create a directory test_dir
go inside it create a test.txt file

Absolute path: Starts from the root directory /

absolute path

```
/home/hamza/linuxx_workshop/test.txt
/etc/systemd/system
```

Relative path: starts from the current working directory

```
linux/test/file.txt
../sc2/trash.txt
```

realpath command gives you the absolute path of a file

```
Hamza@SC2:~/2026/CTF-101/PWN$ realpath ch1
/home/ding/SC2/2026/CTF-101/PWN/ch1
```

users and groups

- A user is an account that can log in, run programs, or own files.
- A group is a collection of user accounts
- When a new user (e.g., hamza) is created, the system automatically creates a group with the exact same name (hamza).

root is a default user in all linux distros with all privileges

root has unlimited system privileges and can access, modify, or delete any file or process.

/etc/passswd: a file storing users information
/etc/shadow: a file storing users passwords

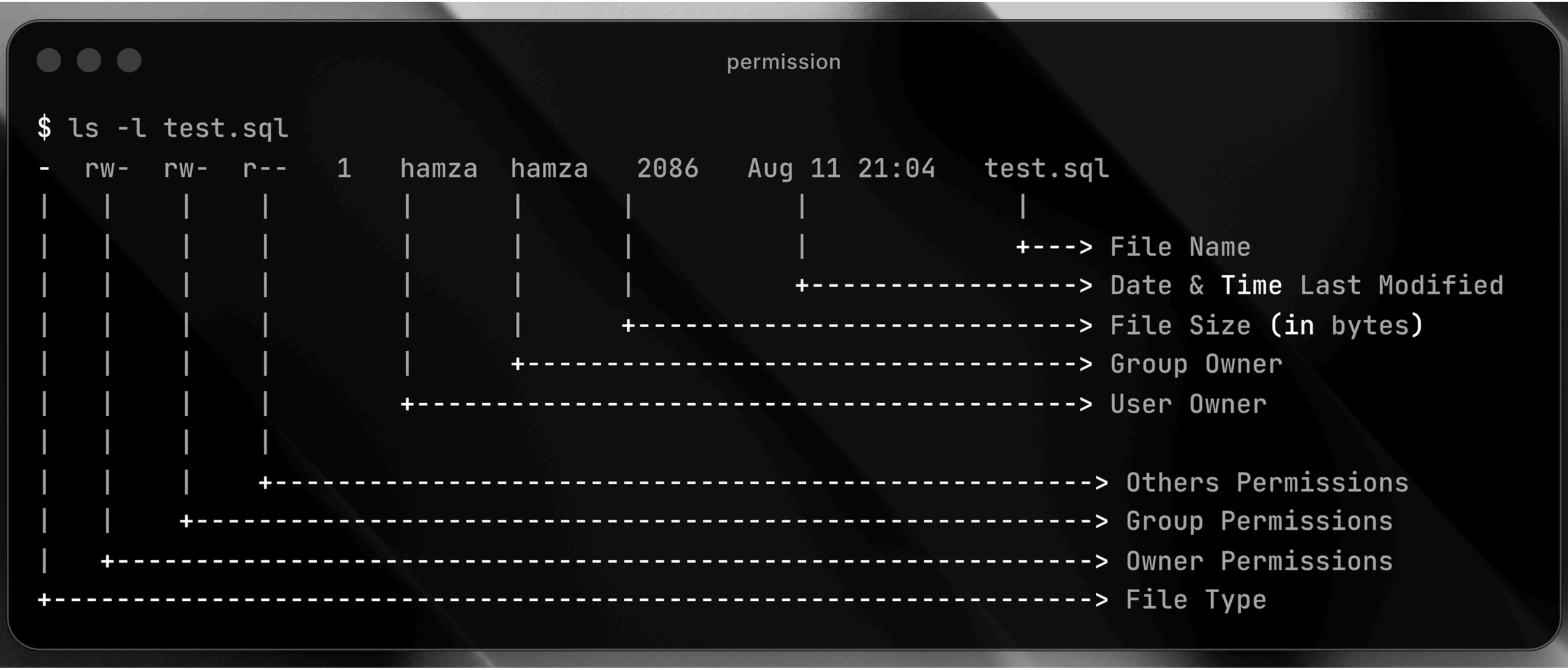
Permissions

r read permission

w write permission

x execute permission

```
rw-rw-r--
rwsr-sr-x
rwxr-xr-w
rwsrwsrwt
```


permission
$ ls -l test.sql
- rw- rw- r-- 1 hamza hamza 2086 Aug 11 21:04 test.sql
File Name
Date & Time Last Modified
File Size (in bytes)
Group Owner
User Owner
Others Permissions
Group Permissions
Owner Permissions
File Type

sudo

gives a normal user user priviliges momentarily

```
>$ sudo
```

Privilege Escalation

Gaining higher access rights and permissions than you are granted

Challenge